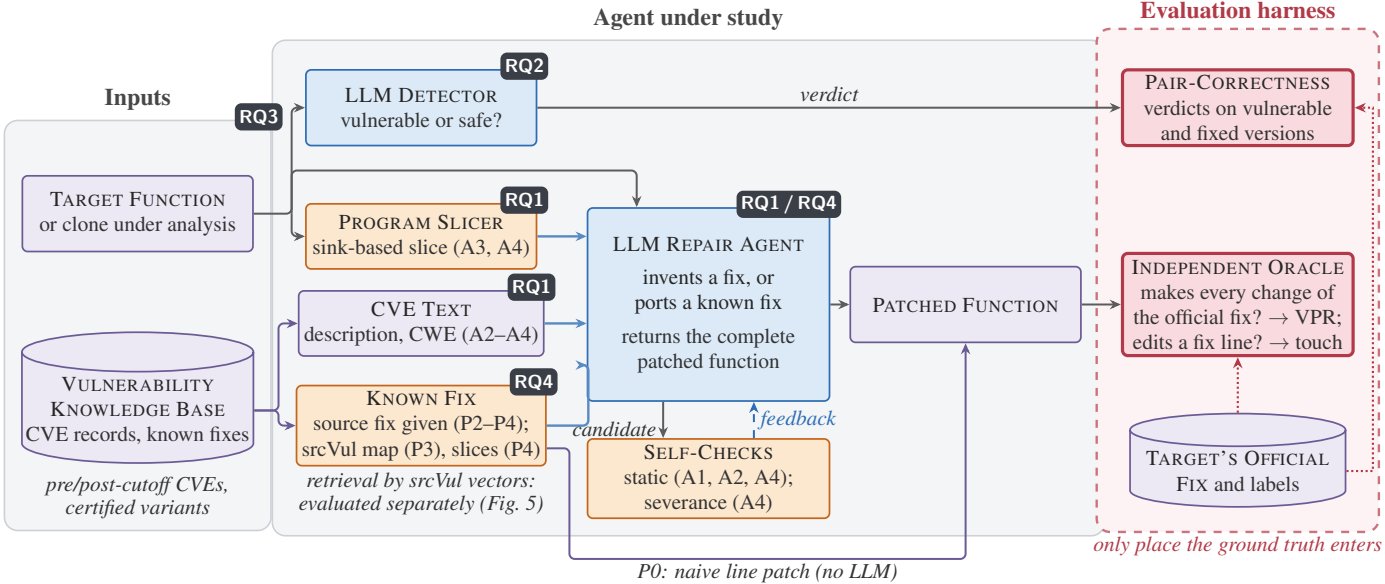

Agent under study
Evaluation harness
Inputs
RQ2
LLM Detector
vulnerable or safe?
verdict
RQ3
Pair-Correctness
verdicts on vulnerable and fixed versions
Target Function
or clone under analysis
RQ1
Program Slicer
sink-based slice (A3, A4)
RQ1 / RQ4
LLM Repair Agent
invents a fix, or ports a known fix
returns the complete patched function
Patched Function
Independent Oracle
makes every change of the official fix? → VPR; edits a fix line? → touch
RQ1
CVE Text
description, CWE (A2–A4)
Vulnerability Knowledge Base
CVE records, known fixes
RQ4
Known Fix
source fix given (P2–P4); srcVul map (P3), slices (P4)
candidate
feedback
Self-Checks
static (A1, A2, A4); severance (A4)
Target's Official Fix and labels
pre/post-cutoff CVEs, certified variants
retrieval by srcVul vectors: evaluated separately (Fig. 5)
only place the ground truth enters
P0: naive line patch (no LLM)
Data & inputs
Slicing / static analysis
LLM component
Evaluation
grounding
feedback
ground truth
RQ probed by research question
A∗, P∗: arms using a component